

# How an LLM writes, one token at a time

## 1 Text becomes tokens

Chunks, not words. “ransomware” became 4 tokens in our toy tokenizer. You pay per token.

## 2 One token at a time

Read everything so far, get a probability for every possible next token, pick one, append it, repeat.

## 3 Temperature sets the randomness

Divide the scores by  $T$  before the softmax. Low  $T$ : predictable. High  $T$ : creative, then nonsense.

## 4 Length is time

Each token waits for the last. With our illustrative numbers: 1 token 0.5 s, 1,500 tokens 25 s.

## 5 Same question, different answers

Sampling means repeat calls can disagree: 39 of 40 borderline alerts got both answers across 20 asks.

## 6 Fluent isn't true

The loop optimises for a plausible next token, not a correct answer. Grounding and checking come from outside.